

PROJECT OVERVIEW

BusinessIQ

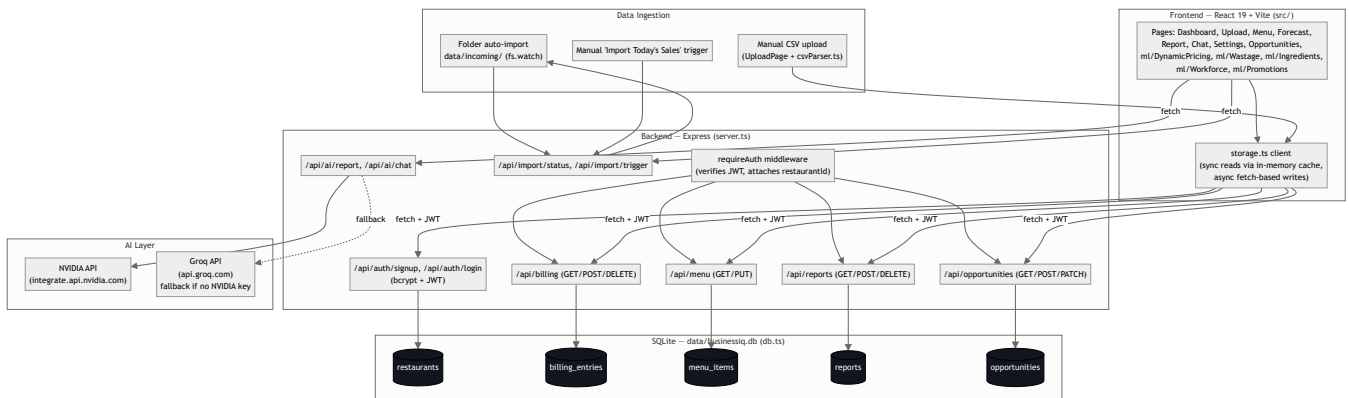
An AI business analyst for micro-restaurants — a bird's-eye view of the system, what's built, and what's next.

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For internal team reference

1. What Is BusinessIQ

BusinessIQ is an AI-powered business analyst built for small, independent ("micro") restaurants. An owner uploads or auto-imports their raw sales data (CSV exports from POS systems, or manual entry), and the app turns that into profitability breakdowns, demand forecasting, dynamic pricing suggestions, food-wastage predictions, ingredient-level demand forecasts, staffing/workforce plans, and promotion-effectiveness analysis — all computed directly from the restaurant's own historical sales, without requiring AI for the core numbers. On top of these individual modules sits an "Opportunity Engine" that proactively watches all of this data together and surfaces the 5 highest-impact, most confident recommendations each week (e.g. "reprice this dish," "this dish is trending down," "stop over-preparing this item") — acting less like a dashboard and more like an analyst pointing things out unprompted. An optional AI layer (NVIDIA or Groq-hosted LLMs) adds natural-language report summaries and a chat interface on top of the same underlying data.

2. System Architecture (current, as verified in code)



Note: restaurants profile fields captured at onboarding (cuisine, establishment type, priorities, etc.) still live in browser localStorage only — not yet in this schema. See Known Gaps.

3. Module Status

Module	Status	Notes
Auth (signup/login/JWT)	Done	bcrypt + JWT, per-account isolation, curl + live-flow verified.
Menu Engineering	Done	buildMenuFromBilling / classifyMenu (Star/Hidden Gem/Volume Trap/Dead Weight), unit-tested.
Forecasting (sales)	Done	Weighted-moving-average backtest engine, unit-tested with hand-calc'd expected values.
Dynamic Pricing	Done	Log-log elasticity / Lerner markup rule, unit-tested.
Wastage Prediction	Done	Newsvendor critical-ratio model, unit-tested.
Ingredient Forecast	Done	Feature-weighted WMA, unit-tested — but see known bug in Section 5 (festival-proximity sentinel).
Workforce Planning	Done	Feature-weighted-vote staffing model, unit-tested.
Promotion Analysis	Done	OLS Interrupted-Time-Series regression, unit-tested.
AI Chat / Reports	In Progress	Code complete (NVIDIA→Groq fallback, isolated failure handling verified live) but no API key was configured on the server as of the last verification — feature is not currently live.
Opportunity Engine	Done	5-signal synthesis layer, Dashboard panel + full history page, verified end-to-end in isolation and live.
Automated Tests	Done	Vitest suite, 69/69 passing across 10 files. Note: /api/menu, /api/reports, /api/opportunities lack dedicated API test cases (only /api/billing has full coverage).
Data Migration to Backend	In Progress	Billing, menu, reports, and opportunities are fully migrated to server-side SQLite with tenant isolation. Restaurant profile (name, cuisine, priorities, etc.) intentionally still browser-only — explicitly flagged as next.

4. Recent Build Timeline

1. **2026-08-27** — Generated Phase-2 project review documentation (CONTENT.md), traced strictly to actual code.
2. **2026-08-27** — Fixed CSV upload to fully replace (not merge) billing/menu data on each upload.
3. **2026-08-27** — Added Groq as a fallback AI provider when NVIDIA's key isn't configured.
4. **2026-08-28** — Hardened CSV upload validation: per-row error/warning detail, ambiguous-date flagging, collapsible Import Issues panel.
5. **2026-08-28** — Built real restaurant accounts: SQLite-backed signup/login, JWT sessions, per-account data namespacing.
6. **2026-08-28** — Added scheduled folder-watch auto-import plus a manual "Import Today's Sales" trigger, sharing one import pipeline.
7. **2026-08-28** — Rebuilt onboarding as a 5-step wizard capturing cuisine/format/tracking-method/priorities; removed sample-data features.
8. **2026-08-28** — Overhauled Reports: real per-date data report that works standalone, with AI commentary isolated so its failures never pollute report history.
9. **2026-08-28** — Upgraded Dashboard visuals: computed insights strip, KPI status colors, Menu Health widget, chart gridlines.
10. **2026-08-28** — Added a Settings page (editable profile + guarded "Danger Zone" full data reset).
11. **2026-08-28** — Added hourly sales, top dishes, meal-period split, and 7-day trend charts to the per-date Report page.
12. **2026-08-28** — Shipped the Opportunity Engine: cross-module signal detection, ranked recommendations, Dashboard panel, and history page.
13. **2026-08-29** — Fixed a data-loss bug where new CSV uploads/auto-imports wiped prior billing history instead of appending to it.
14. **2026-08-29** — Migrated billing, menu, reports, and opportunities from browser localStorage to server-side SQLite with real tenant isolation, so data survives a cleared browser or a new device.
15. **2026-08-29** — Added an automated test suite (69 checks) covering every prediction engine, the menu logic, the Opportunity Engine, and the login/API layer — including recovering a real dataset that an early test-setup bug accidentally deleted.

5. Known Gaps / Next Up

Restaurant profile still browser-only. Name, owner, city, cuisine, establishment type, and priorities were intentionally left out of the server migration and still live only in localStorage — explicitly logged as "next on the list."

Two unresolved ML weighting inconsistencies (found while writing tests, deliberately left as product decisions, not silently fixed): (1) `features.ts`'s `festivalProximity()` "nothing nearby" sentinel value of 7 incorrectly satisfies `ingredientForecast.ts`'s `<=7` check, giving every far-from-festival forecast an unwarranted +8% demand bump. (2) `wastagePredictor.ts` and `ingredientForecast.ts` weight same-weekday history oldest-first, the reverse of `forecasting.ts`'s own (correct) most-recent-first weighting.

Dashboard "Revenue by Meal Period" donut renders zero pie sectors. Confirmed via DOM inspection, pre-existing, flagged twice across two separate tasks, still unfixed.

API test coverage is partial. `/api/menu`, `/api/reports`, and `/api/opportunities` exist and work but aren't covered by dedicated test cases — only `/api/billing` has full tenant-isolation + dedupe test coverage. Flagged as a follow-up extension.

AI features need a live API key. AI report/chat code paths are complete and their failure states verified, but no `NVIDIA_API_KEY` or `GROQ_API_KEY` was configured on the server as of the last verification — so AI Chat/Reports are not currently live for users.

Import-path personalization deferred by design. The onboarding wizard now captures each restaurant's tracking method and POS name, but using that to reorder/personalize the Upload page's import options (and AI tone) was explicitly scoped out of that task and not yet built.

